

FMECA Studio

User Guide for Reliability Engineers

Building professional FMECA tables for RCM projects

Corporate (Cloudflare) Deployment • Microsoft Copilot via Power Automate

Worked example: Thulailat Instrument-Air Compressor Package

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1. Introduction

FMECA Studio is a browser-based workbench for building Failure Mode, Effects and Criticality Analysis (FMECA) worksheets as part of a Reliability-Centered Maintenance (RCM) program. It guides you from a blank project to a complete, exportable FMECA table, and uses an AI assistant to accelerate the repetitive drafting work while keeping the reliability engineer firmly in control of every judgement.

The application is fully self-contained: your projects are stored locally in your browser, and analytical content is generated on demand through your organization's approved AI channel. There is no separate database to install and nothing leaves your machine except the specific prompts you choose to send to the AI service.

1.1 Who this guide is for

This guide is written for reliability, maintenance and integrity engineers who are responsible for producing FMECA / FMEA worksheets, criticality assessments and maintenance task selections. It assumes you understand the reliability concepts (functions, functional failures, failure modes, effects, causes and RPN) but not the tool. Every feature is illustrated with the Thulailat Instrument-Air (IA) Compressor Package example project.

1.2 How FMECA Studio structures an analysis

The tool follows the classic RCM decomposition. Understanding this hierarchy is the single most important thing before you start — every screen in the application is organised around it.

Level	Meaning	Thulailat example
Project	The asset / package under analysis	Thulailat IA Compressor
Subsystem	A maintainable sub-assembly with its own specs & function	Screw Compressor; Lubrication Oil System
Function	What the subsystem must do, with performance standards	Compress air to instrument quality; deliver 600 SCFM at 10 barg
Functional Failure	A way the function is not delivered	Produces non-instrument-quality air
Failure Mode	The specific mechanism that causes the failure	Oil/air separator element saturation / breakthrough
Effect / Cause	Local & end consequences, and the root cause	Oil carry-over to dryer; element service-life exceeded
Controls & Mitigation	Existing controls and recommended actions (with owner)	Separator ΔP alarm; run-hour-based replacement
RPN	Severity × Occurrence × Detection risk score	$S 5 \times O 5 \times D 2 = 50$

Data model at a glance

Project → Subsystems → Functions → Functional Failures → Failure Modes → (Effect, Cause, Controls, Mitigation, RPN)

2. Getting started

2.1 Opening the application

FMECA Studio runs entirely in your web browser — there is nothing to install. Open the corporate deployment URL provided by your reliability team (the application is hosted on Cloudflare). It loads directly to the Dashboard shown in the next section. Because your work is saved in the browser's local storage, always use the same browser profile on the same machine to return to your projects, and export a JSON backup regularly (Section 7).

Good to know

Projects auto-save continuously — the header shows a “● Auto-saved” indicator. There is no manual “Save” button. Clearing your browser data will delete unsaved projects, so keep JSON exports of anything important.

2.2 Connecting the AI provider (corporate Copilot)

Before the AI features will work you must point FMECA Studio at your organization's AI channel. Open Settings from the Dashboard (the “Settings” pill under the title). In the corporate deployment the AI Provider panel deliberately exposes only the approved options — Copilot and OpenRouter — so that analysts cannot route company data through personal API keys.

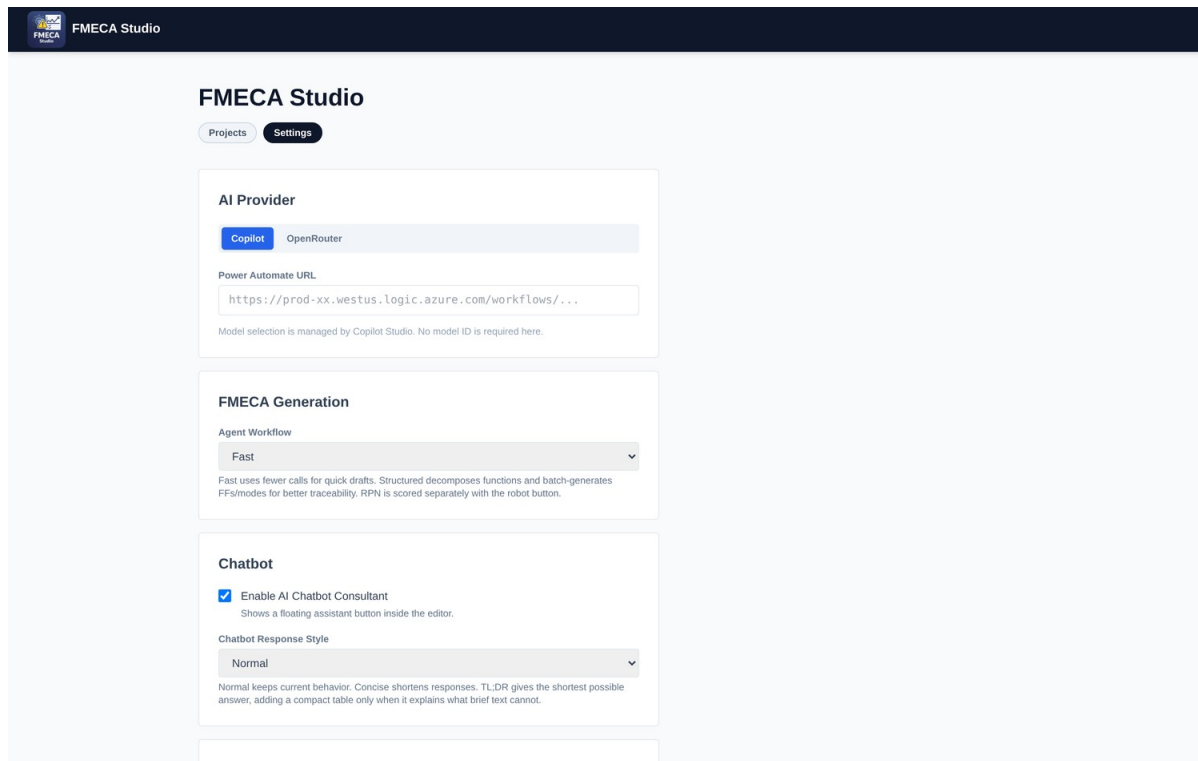


Figure 2.1 — The AI Provider panel in the corporate build. Only Copilot and OpenRouter are shown.

For corporate use, keep the provider on **Copilot** and paste your **Power Automate URL** into the field provided. This is the HTTP-trigger endpoint of the Power Automate flow that fronts your Microsoft Copilot Studio agent (it looks like `https://prod-xx...logic.azure.com/workflows/...`). No API key and no model name are required — model selection is managed centrally by Copilot Studio.

Corporate policy

Company reliability data must only be processed through the approved Copilot channel. Do not switch the provider away from Copilot, and do not paste personal API keys. If the Power Automate URL is blank, all AI buttons (Auto-Fill, Generate, the RPN robot and the Copilot chat) will prompt you that the URL is required.

2.3 Other useful settings

Three further settings shape how the AI behaves. You can return to change them at any time.

- **Agent Workflow (FMECA Generation).** “Fast” uses fewer calls for quick drafts; “Structured” decomposes functions and batch-generates functional failures and modes for better traceability. RPN is always scored separately with the robot button.
- **Chatbot.** Enables the floating “FMECA Consultant” assistant inside the editor and sets its response style (Normal, Concise or TL;DR).
- **AI Source.** Chooses what context the AI draws on: “AI only” (general knowledge), “File only (RAG)” (a reference document you upload) or “Hybrid” (your project data plus the reference file). You can also load a Global Reference File and a Checklist Knowledge file (PM checklists by team and interval) that feed mitigation suggestions.

3. The Dashboard

The Dashboard is the home screen. It has two tabs — Projects and Settings. The Projects tab lists every analysis stored in your browser as a card showing its name, description and created / updated dates.

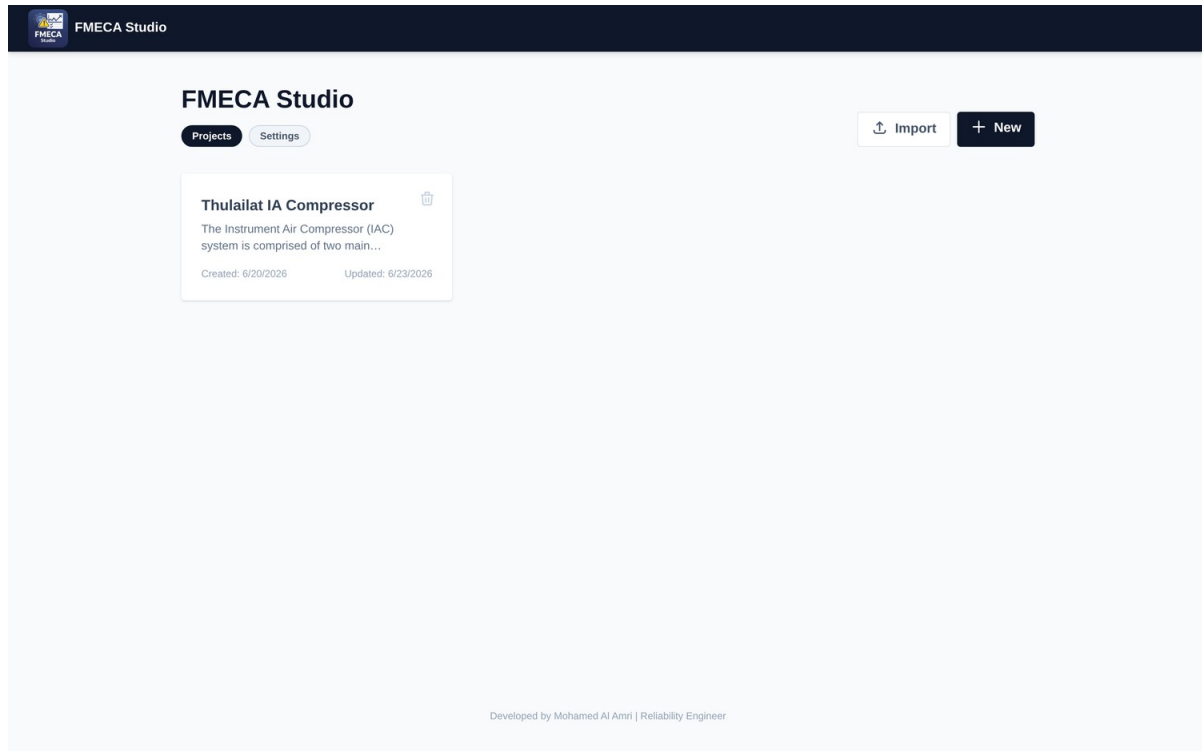


Figure 3.1 — The Projects dashboard with the Thulailat IA Compressor project.

From here you can:

- **Create a new project** — click New to start a blank analysis and open the editor.
- **Import a project** — click Import and select a previously exported FMECA JSON file (this is how you would open the Thulailat example file).
- **Open a project** — click anywhere on a project card to open it in the editor.
- **Delete a project** — use the trash icon on the card (this cannot be undone, so export first).

Starting from the example

To follow this guide hands-on, click Import and choose the file “FMECA_Thulailat IA Compressor.json”. The fully worked analysis will appear as a new card. Click it to explore every feature described below.

4. Building a FMECA step by step

This is the core workflow. Opening a project takes you to the Build view — a hierarchical editor where you expand the asset from the top down. The screenshot below shows the editor for the Thulailat Screw Compressor subsystem, with the project context, subsystem card, function, and the first functional failure and mode.

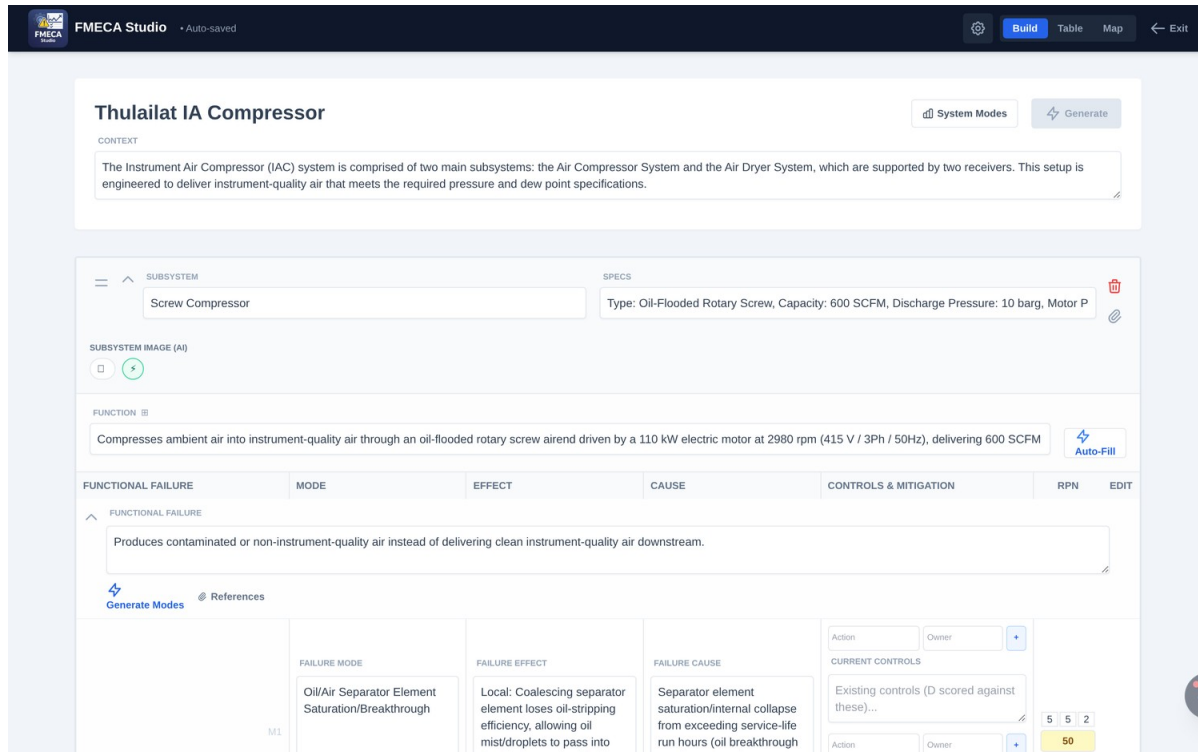


Figure 4.1 — The Build editor: project context, subsystem, function, functional failure and failure mode with RPN.

4.1 Define the project and context

At the top of the editor, give the project a clear name and write a short Context describing the asset and its operating envelope. In the example, the context explains that the IA package comprises an air-compressor system and an air-dryer system feeding two receivers to deliver instrument-quality air. This context is important: it is fed to the AI as background for every generation, so a good description improves every downstream suggestion.

4.2 Generating the FMECA with AI

FMECA Studio can draft the analysis for you at four levels of granularity — from the whole worksheet in one click down to a single field. You choose how much to automate; every result is fully editable afterwards. The four generation tools are:

Tool	Where it is	What it produces
“Generate” (master)	Top-right of the editor, beside System Modes	The entire FMECA — subsystems, functions, functional failures and failure modes for the whole asset
“Auto-Fill”	Beside each subsystem’s Function field	All functional failures and their modes (mode, effect, cause and mitigation) for that one

Tool	Where it is	What it produces
		subsystem
“Generate Modes”	Under each functional failure	A set of failure-mode rows for that failure — each row filled with mode, effect, cause and mitigation
Magic wand	Inside every text field	Content for that single field — writes it if empty, or fixes typos / grammar on short text

The master “Generate” button — build the whole FMECA at once

This is the fastest way to start. On a new, empty project, fill in the project name and a rich Context, then click the blue **Generate** button. The tool identifies the subsystems, decomposes each function, and creates the functional failures and failure modes for the entire asset, showing a live progress bar (you can click **Cancel** at any point). When it finishes you have a complete draft to review and refine.

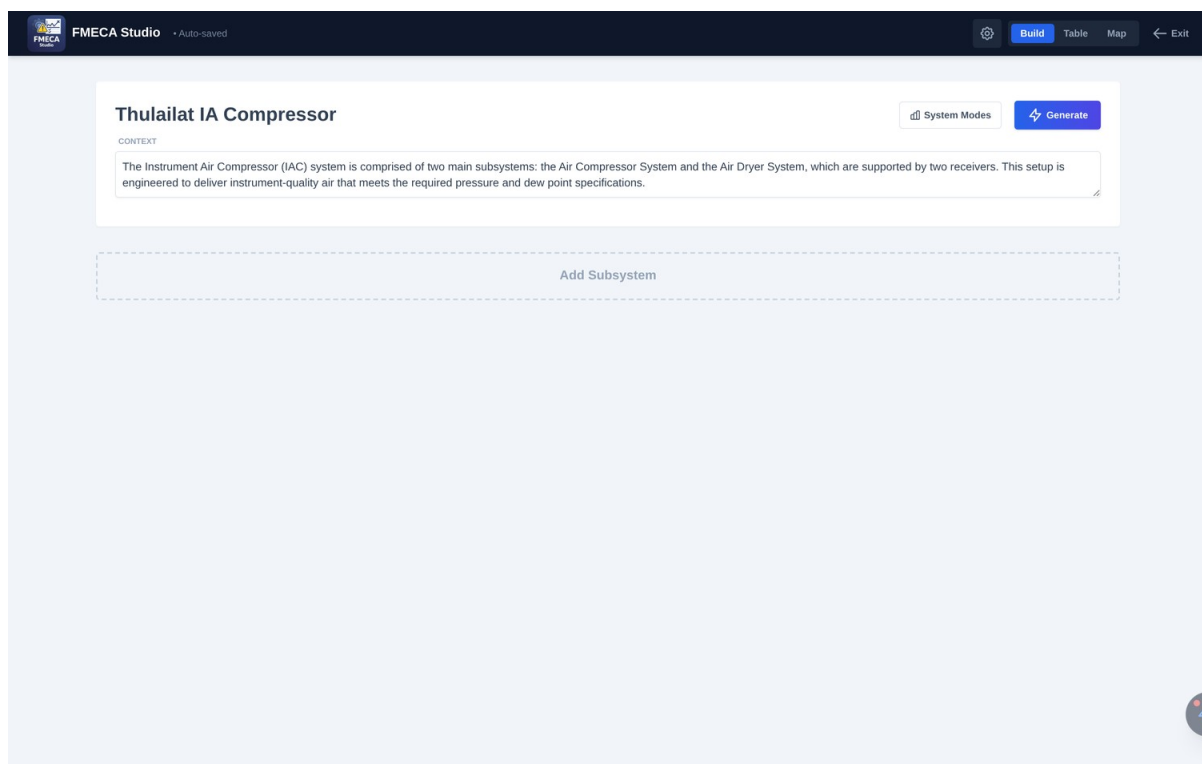


Figure 4.2 — On an empty project the master “Generate” button (top-right) is active and builds the whole FMECA from the project context.

“Generate” works only on an empty FMECA

The master Generate button is available only while the project has no subsystems yet — it is greyed out (“Available only when the FMECA is empty”) once content exists, so it can never overwrite your work. To regenerate from scratch, delete the subsystems first. To add to an existing analysis, use Auto-Fill and Generate Modes instead.

“Auto-Fill” — draft one subsystem

Once a subsystem has a name, specs and a function, click **Auto-Fill** (the wand button next to the Function field) to generate that subsystem’s complete failure analysis — its functional failures and, under each, the failure modes with their effects, causes and mitigations (both the current-controls context and the suggested

mitigation actions). Use this to fill out subsystems one at a time, or to add analysis to a subsystem you created manually.

“Generate Modes” — draft the rows under one failure

On any functional failure, click the **Generate Modes** link (the lightning icon). The AI proposes several credible failure-mode rows for that specific failure, each already populated with the **mode, effect, cause and mitigation**. This is the right tool when you have written the functional failures yourself and only want the modes drafted.

The magic wand — one field at a time

Every text field (Function, Specs, functional-failure description, mode, effect, cause, mitigation) carries a small *magic-wand* icon that appears on hover. It is context-aware and behaves according to how much text is already there:

- **Empty field** — writes appropriate content from scratch, using the project and subsystem context.
- **Short text (a few words)** — corrects spelling and grammar only, leaving your wording intact.
- **Longer text** — enhances and expands what you have already written, keeping your meaning.

Recommended generation workflow

New asset: click **Generate** for a full first draft → review.

Adding to it: use **Auto-Fill** per subsystem or **Generate Modes** per failure.

Polishing: use the **magic wand** on individual fields. Then score RPN (Section 4.7). Always review AI output against plant history.

4.3 Add subsystems, specifications and function

If you prefer to build manually (or need to add to a generated draft), use Add Subsystem to create each maintainable sub-assembly, then fill three fields:

1. **Subsystem name** — e.g. “Screw Compressor”.
2. **Specs** — the nameplate / design data (type, capacity, pressure, motor rating, configuration). The example lists “Oil-Flooded Rotary Screw, 600 SCFM, 10 barg, 110 kW, 2980 rpm, 1 Active + 1 Standby”.
3. **Function** — a full statement of what the subsystem must do and to what standard. Write this carefully; it is the anchor for the whole failure analysis below it.

You do not have to write everything by hand. The **Auto-Fill** button (beside the Function field) asks the AI to draft the function from the subsystem name and specs, and the small *magic-wand* icon inside any text field will generate text from scratch when the field is empty, fix grammar on short text, or enhance longer text you have already written.

4.4 Decompose the function (Function Breakdown)

A single function statement usually bundles several distinct performance requirements. Click the  icon next to the FUNCTION label to open the **Function Breakdown** tool. It splits the function into discrete function / expectation pairs and lets you verify that each one is covered by a functional failure — the backbone of RCM traceability.

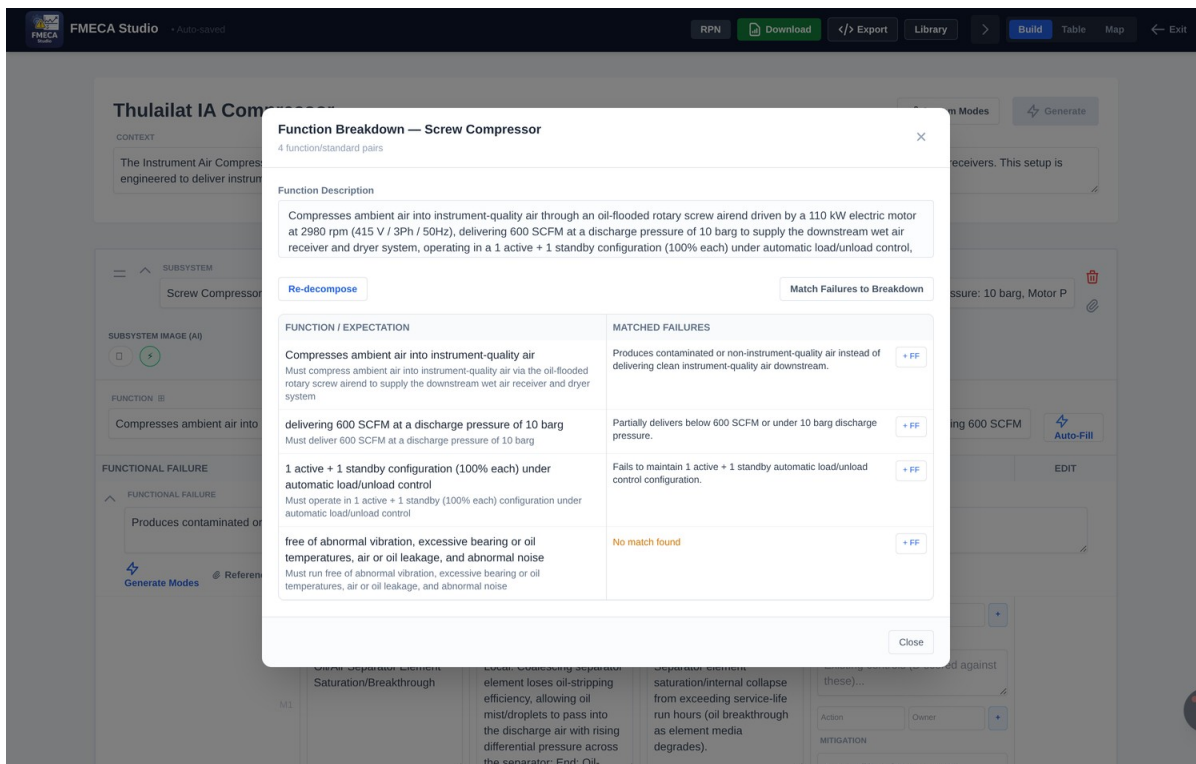


Figure 4.3 — Function Breakdown: each performance standard is matched to a functional failure. “No match found” flags a gap.

Use **Re-decompose** to regenerate the breakdown after editing the function, and **Match Failures to Breakdown** to auto-link existing functional failures to each standard. Where the tool shows “No match found” (as for the abnormal-vibration standard in the example), you have identified a missing functional failure — click **+ FF** to create it. This is a powerful completeness check before you sign off an analysis.

4.5 Add functional failures

Under each subsystem, use **+ Add Failure** to record each way the function can fail, phrased as a loss of function (e.g. “Produces contaminated or non-instrument-quality air instead of delivering clean instrument-quality air downstream”). Keep functional failures at the level of the performance standard, not the mechanism — mechanisms belong in the failure modes below.

4.6 Add failure modes, effects, causes and controls

For each functional failure, add the specific **failure modes** that cause it. Each mode row captures four analytical fields plus mitigation:

- **Failure Mode** — the mechanism (e.g. “Oil/Air separator element saturation / breakthrough”).
- **Failure Effect** — written as Local and End effects (e.g. separator loses efficiency → oil-laden air reaches the dryer and instruments).
- **Failure Cause** — the underlying reason (e.g. element service-life exceeded).
- **Current Controls** — barriers already in place; the Detection score is judged against these.
- **Mitigation** — recommended actions. The Mitigation builder lets you enter each action with an Owner and auto-numbers the list, so tasks read as “1- ... (Mechanical team); 2- ... (Instrument team)”.

The **Generate Modes** link (the lightning icon on a functional failure) asks the AI to propose a set of credible modes with their effects, causes and mitigations, which you then review and edit. Treat AI output as a first draft to be checked against your plant history — never as the final word.

4.7 Score the risk (RPN)

FMECA Studio uses the standard Risk Priority Number: **RPN = Severity × Occurrence × Detection**, each rated on a 1–10 scale. Enter the three numbers in the S / O / D boxes on each mode row, or click the **AI-score (robot) button** on the row to have the AI propose a defensible S/O/D based on the effect, cause and current controls. The computed RPN is colour-coded so the worst risks stand out at a glance:

RPN band	Colour in the app	Interpretation
RPN ≥ 100	Red	High priority — act first; strengthen detection / prevention
40 – 99	Yellow	Medium — plan mitigation
Below 40	Green	Low — monitor
Blank / Unscored	Grey	Not yet assessed

Baseline vs. mitigated RPN

The AI scorer can record a baseline RPN (using only the current controls) alongside the mitigated RPN (after the recommended actions), so you can demonstrate the risk reduction your maintenance strategy delivers. Turn the RPN column on or off with the RPN toggle in the toolbar.

5. AI-assisted features

Beyond the per-field wand and the Generate buttons, two features help you bring real operating knowledge and a reference library into the analysis.

5.1 System Modes — injecting operational failure data

The **System Modes** button (top-right of the editor) lets you upload real failure history — an Excel sheet with two columns, Failure Mode and Failure Count (occurrences). FMECA Studio ranks this data and injects it as context into every AI prompt, so generated modes reflect what actually fails on your equipment rather than generic textbook modes.

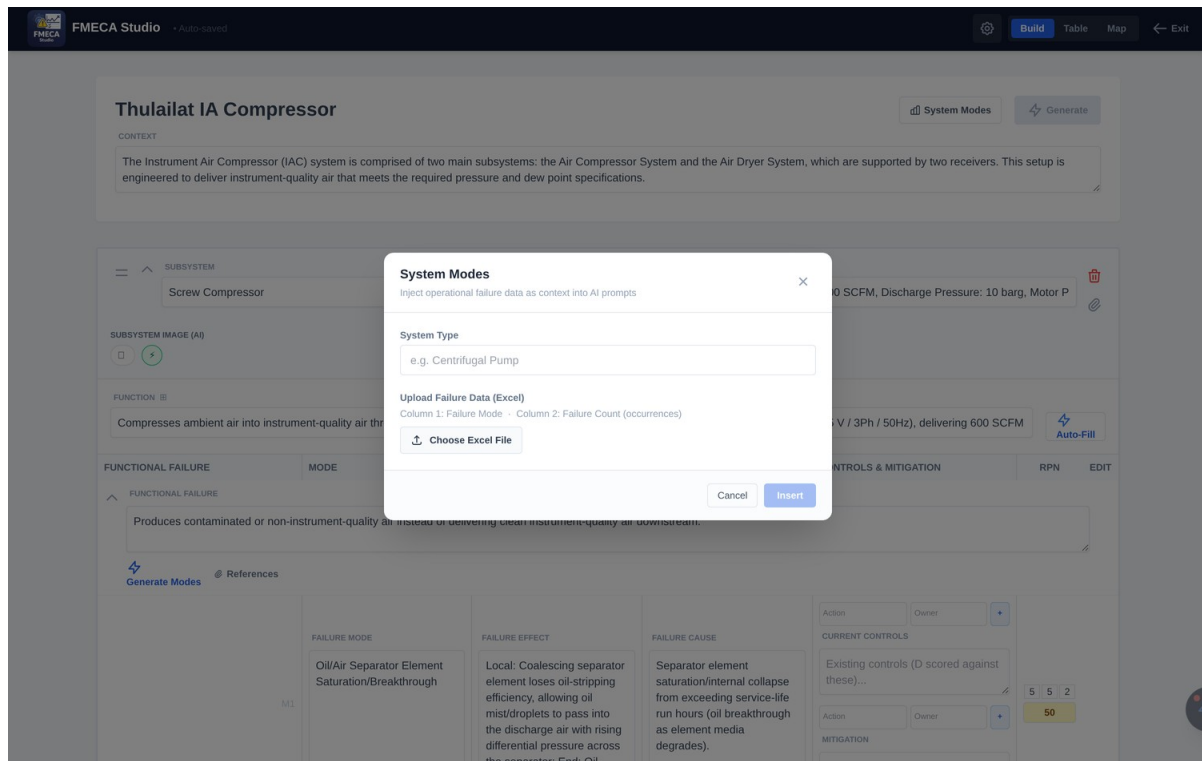


Figure 5.1 — System Modes: upload ranked operational failure counts to ground the AI in real plant history.

5.2 The failure library

The **Library** panel (open the toolbar, then click Library) holds reusable failure records grouped by equipment class — for example Centrifugal Pumps and Electric Motors — each with a Mode, Effect, Cause and suggested Task. Use it as a starting palette for common equipment, and load your own library with the upload icon (download the template first with “Get Template”).

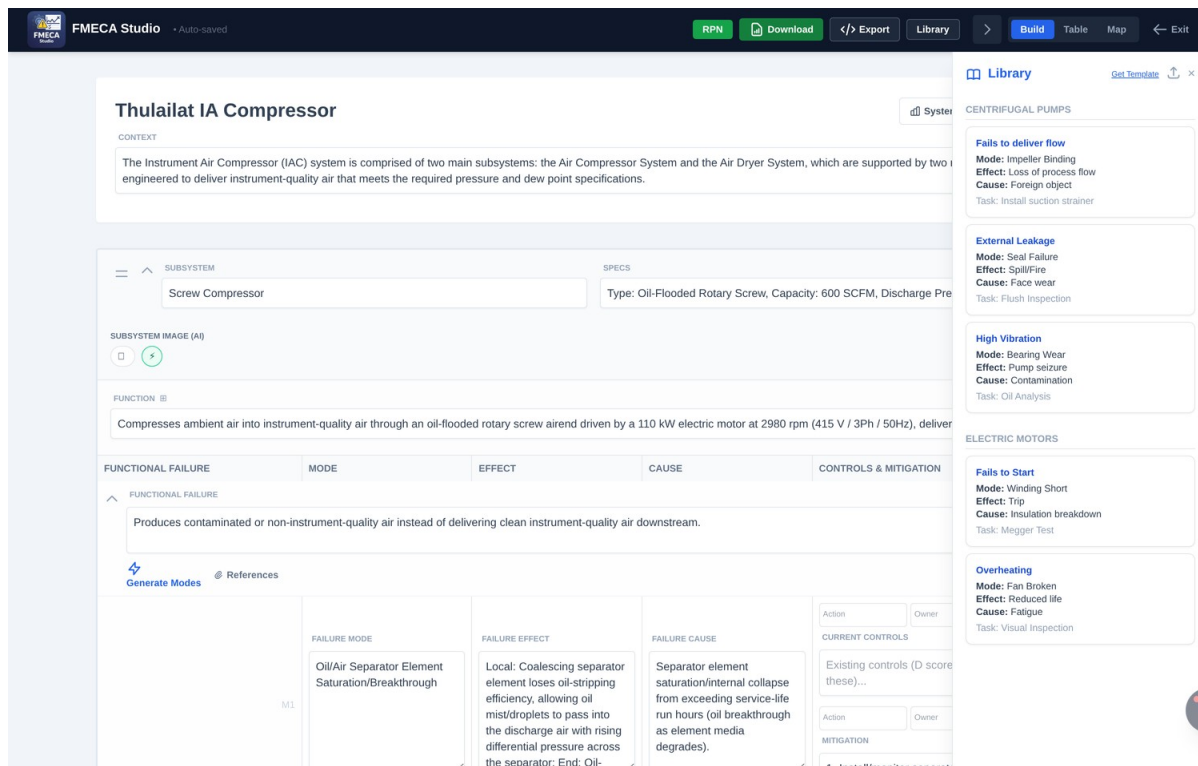


Figure 5.2 — The failure library, grouped by equipment type.

5.3 The FMECA Copilot chatbot

The floating **FMECA Consultant** (the wand tab on the right edge of the editor) is a senior-reliability-engineer chat assistant. With **RAG ON** it retrieves exact data from your open project — subsystems, failure modes and RPN summaries — before answering, so you can ask questions such as “Which modes have the highest RPN?” or “Suggest additional detection for the oil-carryover mode.” The green “Project Context Active” badge confirms it is reading your live data.

The screenshot displays the FMECA Studio software interface. At the top, the title bar reads "FMECA Studio" with a status indicator "Auto-saved" and navigation buttons "Build", "Table", "Map", and "Exit". The main workspace is titled "Thulailat IA Compressor".

CONTEXT

The Instrument Air Compressor (IAC) system is comprised of two main subsystems: the Air Compressor System and the Air Dryer System, which are engineered to deliver instrument-quality air that meets the required pressure and dew point specifications.

SUBSYSTEM

Screw Compressor

SPECS

Type: Oil-Flooded Rotary Screw, Capacity: 600 SCFM

SUBSYSTEM IMAGE (AI)

FUNCTION

Compresses ambient air into instrument-quality air through an oil-flooded rotary screw airend driven by a 110 kW electric motor at 2980 rpm (415 V / 3PH)

FUNCTIONAL FAILURE

FUNCTIONAL FAILURE	MODE	EFFECT	CAUSE	CONTROLS
Produces contaminated or non-instrument-quality air instead of delivering clean instrument-quality air downstream.				

Generate Modes @ References

The chatbot overlay, titled "FMECA Consultant", is active and displays the message: "Hello! I am your FMECA Consultant. I can help analyze your FMECA project, suggest improvements, or identify risks. How can I assist you today?". Below the chatbot is an input field with the placeholder text "Ask about risks, mitigations..." and a send button.

The background interface shows a table with columns for FAILURE MODE, FAILURE EFFECT, FAILURE CAUSE, and CURRENT CONTROLS. The FAILURE MODE column contains the text "Oil/Air Separator Element Saturation/Breakthrough". The FAILURE EFFECT column contains the text "Local: Coalescing separator element loses oil-stripping efficiency, allowing oil mist/droplets to pass into the discharge air with rising differential pressure across the separator; End: Oil-". The FAILURE CAUSE column contains the text "Separator element saturation/internal collapse from exceeding service-life run hours (oil breakthrough as element media degrades)". The CURRENT CONTROLS column contains the text "Existing controls (D scored against these)..." and a table with the values 5, 5, and 2, and a score of 50.

Figure 5.3 — The FMECA Consultant chatbot with retrieval (RAG) over the open project.

6. Views and the toolbar

The editor offers three views of the same data, switched from the Build / Table / Map buttons in the header. The toolbar (the gear/arrow button) reveals the RPN toggle and the export buttons.

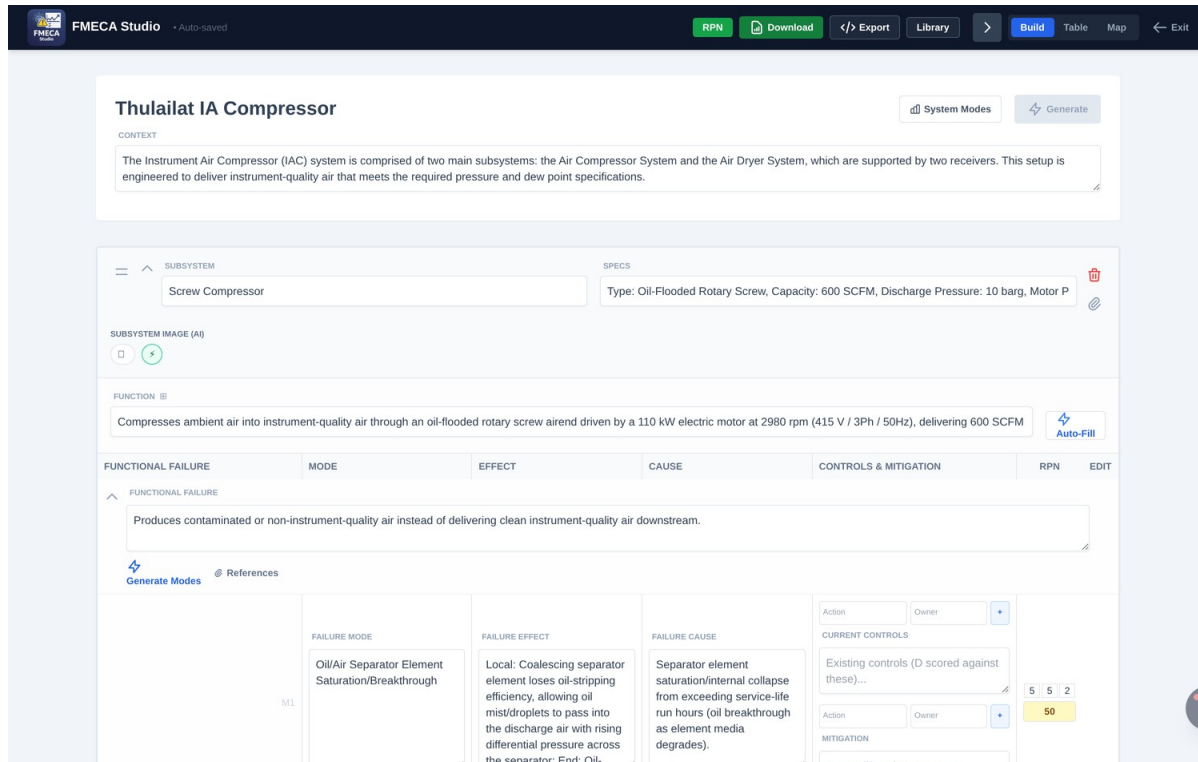


Figure 6.1 — The expanded toolbar: RPN toggle, Excel Download, JSON Export and Library.

6.1 Table view

The Table view renders the analysis as a classic spreadsheet-style FMECA worksheet, with subsystem and functional-failure cells merged across their modes and the S / O / D / RPN columns on the right. This is the review-and-sign-off view and the layout that matches the exported Excel file.

SUBSYSTEM	SPECS	FUNCTION	FUNCTIONAL FAILURE	FAILURE MODE	FAILURE EFFECT	FAILURE CAUSE	CURRENT CONTROLS	MITIGATION	S	O	D	RPN
Screw Compressor	Type: Oil-Flooded Rotary Screw, Capacity: 600 SCFM, Discharge Pressure: 10 barg, Motor Power: 110 kW, Motor Speed: 2980 rpm, Voltage: 415 V / 3Ph / 50Hz, Configuration: 1 Active + 1 Standby (100% each)	Compresses ambient air into instrument-quality air through an oil-flooded rotary screw arend driven by a 110 kW electric motor at 2980 rpm (415 V / 3Ph / 50Hz), delivering 600 SCFM at a discharge pressure of 10 barg to supply the downstream wet air receiver and dryer system, operating in a 1 active + 1 standby configuration (100% each) under automatic load/unload control, while running free of abnormal vibration, excessive bearing or oil temperatures, air or oil leakage, and abnormal noise.	Produces contaminated or non-instrument-quality air instead of delivering clean instrument-quality air downstream.	Oil/Air Separator Element Saturation/Breakthrough	Local: Coalescing separator element loses oil-stripping efficiency, allowing oil mist/droplets to pass into the discharge air with rising differential pressure across the separator; End: Oil-laden air is delivered to the downstream wet air receiver and dryer, contaminating instrument-quality air and risking fouling of dryer desiccant and instrument valves/sensors.	Separator element saturation/internal collapse from exceeding service-life run hours (oil breakthrough as element media degrades).		1- Install/monitor separator differential-pressure transmitter and alarm to detect element saturation early [Tag: PDT-SEP (Hi: 0.8 barg, Hi-Hi: 1.0 barg)] (Instrument team) 2- Implement run-hour-based separator element replacement at 4,000-8,000 hr or on high DP, tracking operating hours rather than visual inspection (Mechanical team) 3- Add downstream oil-content / hydrocarbon monitoring at the receiver inlet to detect oil breakthrough before it reaches instruments [Tag: AT-Oil] (Instrument team) 4- Perform periodic oil analysis to confirm correct fluid type/viscosity and detect foaming tendency that accelerates separator saturation (Mechanical team)	5	5	2	50
				Oil Scavenge/Return Line Restriction (Oil Carryover)	Local: Separated oil accumulates in the separator housing because the scavenge/return line is blocked, kinked, or sludged, causing oil level to rise to the bottom of the separator element; End: Excess oil exits with the air stream into the downstream receiver and dryer, producing oil-contaminated, non-instrument-quality air.	Scavenge/oil-return line blockage or restriction (sludge buildup, blocked scavenge filter/orifice, or kinked line) preventing return of separated oil.		1- Schedule periodic cleaning/inspection of scavenge line, scavenge strainer/orifice, and return line for sludge, kinks, and correct angle/clearance (Mechanical team) 2- Use correct-viscosity, low-sludge lubricant and enforce oil-change intervals to prevent sludge-driven scavenge clogging (Mechanical team)				Unscored

Figure 6.2 — Table view: the structured FMECA worksheet with merged cells and RPN.

6.2 Map view

The Map view draws the analysis as a top-down tree — project → subsystems → functional failures → modes. Use **Expand** / **Collapse**, the zoom control and the level selector (Subsystems / deeper levels) to focus the diagram.

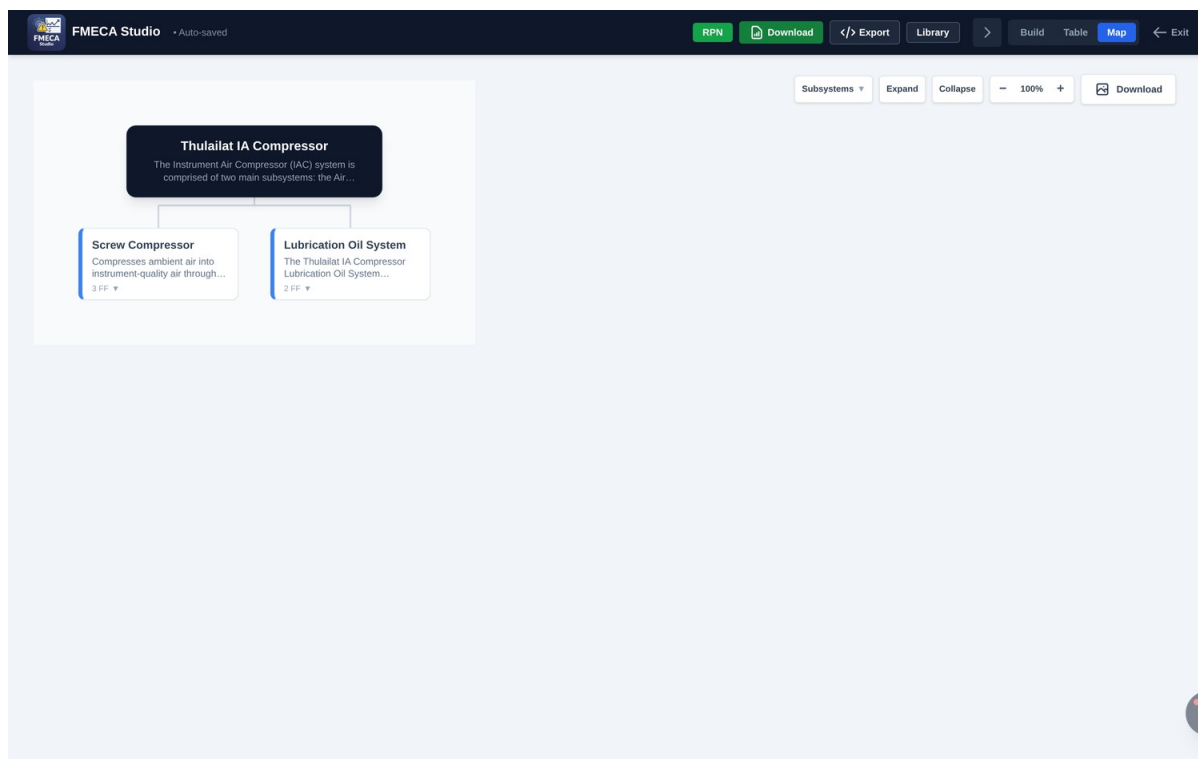


Figure 6.3 — Map view: a navigable tree of the asset breakdown.

Downloading the map

Click the **Download** button (top-right of the Map) to open a menu with four export formats. Choose the one that suits the destination:

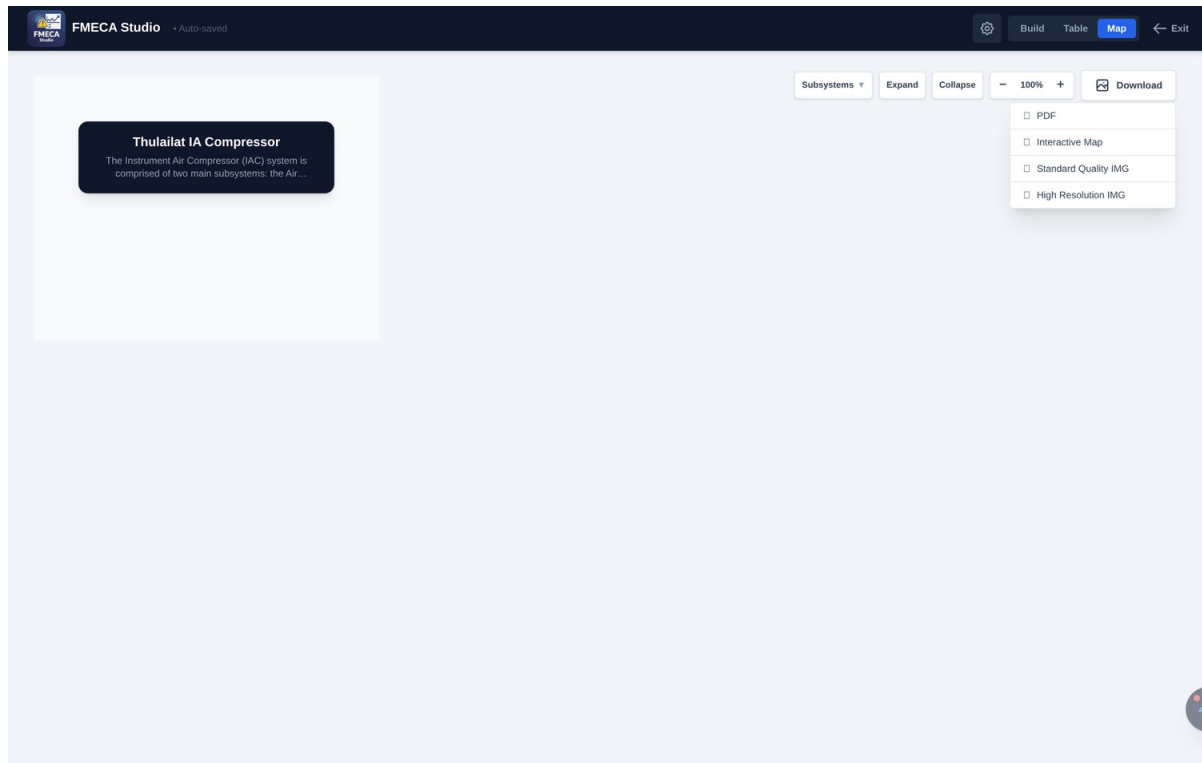


Figure 6.4 — The Map Download menu offers PDF, an interactive HTML map, and two PNG resolutions.

Menu option	File type	Best for
PDF	PDF	A print-ready page of the map for reports, sign-off packs and archiving
Interactive Map	HTML	A self-contained web page you can open in a browser and pan / expand — ideal for sharing a clickable map by email
Standard Quality IMG	PNG	A quick, screen-resolution image for slides and documents
High Resolution IMG	PNG	A high-resolution (4×) image for large-format printing or zooming in

Tip

The PDF and HTML exports capture the map exactly as it is currently expanded and zoomed, so set the level selector and Expand / Collapse the way you want before exporting.

7. Exporting and sharing

FMECA Studio produces several export formats from the toolbar and the Map view:

Format	Where	Use it for
Excel (.xlsx)	Download (toolbar)	The formatted FMECA worksheet with merged cells — the main deliverable for review and archiving
JSON	Export (toolbar)	A complete backup of the project; re-import it on any machine or share with a colleague
PDF	Download menu (Map view)	A print-ready page of the asset-breakdown tree
HTML (Interactive Map)	Download menu (Map view)	A shareable, browser-openable interactive map
PNG (Standard / High-Res)	Download menu (Map view)	The tree diagram as an image for slides, documents and printing

Back up regularly

Because projects live in your browser's local storage, export a JSON file at the end of every working session and store it with your project records. This is your only backup if the browser data is cleared.

8. Best practice for a professional FMECA

The tool accelerates the work, but a defensible FMECA still depends on engineering discipline. The following practices produce a worksheet that will stand up to audit.

- **Write functions to a measurable standard.** “Deliver 600 SCFM at 10 barg” is auditable; “supply air” is not. Strong function statements make every failure below them sharper.
- **Close every gap in the Function Breakdown.** Treat each “No match found” as an action to add a functional failure before sign-off.
- **Ground the AI in real data.** Load operating history through System Modes and a PM checklist through Checklist Knowledge so mitigations reflect your plant, not a generic template.
- **Always review AI output.** Generated modes, effects and RPN scores are drafts. Confirm them against plant history and your own judgement — you remain the author of record.
- **Score detection against real controls.** Fill Current Controls before scoring D, and use the baseline-vs-mitigated RPN to show the value of your recommendations.
- **Name an owner for every action.** The Mitigation builder’s Owner field turns recommendations into assignable tasks.
- **Export a JSON backup and an Excel deliverable at each milestone.** Version them with the analysis date.

Quick-reference: end-to-end workflow

1. Import or create a project and write a clear Context.
2. Add each subsystem with its specs and function (use Auto-Fill to draft).
3. Run Function Breakdown and close any unmatched standards.
4. Add functional failures, then generate and refine failure modes.
5. Complete effects, causes, current controls and mitigations (with owners).
6. Score S/O/D (manually or with the robot); review the coloured RPNs.
7. Check the Table view; ask the Copilot to sanity-check the highest risks.
8. Export Excel for the deliverable and JSON for backup.

— End of guide —